

ASE6029 Linear Optimal Control Homework #3

- 1) *Shift matrix.* Consider the $n \times n$ upper shift matrix,

$$N = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

with

$$J = \lambda I + N = \begin{bmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & \lambda \end{bmatrix}.$$

- a) Find e^{tN} .
- b) Find e^{tJ} . *Hint: Note that λI and N commute.*

- 2) *Jordan form of a block matrix.* We consider the block 2×2 matrix

$$C = \begin{bmatrix} A & I \\ 0 & A \end{bmatrix}.$$

Here $A \in \mathbb{R}^{n \times n}$, and is diagonalizable, with real, distinct eigenvalues $\lambda_1, \dots, \lambda_n$. We'll let $v_1, \dots, v_n$ denote (independent) eigenvectors of A associated with $\lambda_1, \dots, \lambda_n$.

- a) Find the Jordan form J of C . Be sure to explicitly describe its block sizes.
- b) Find a matrix T such that $J = T^{-1}CT$.

- 3) *Linear Independence of a Jordan Chain.* Suppose vectors $v_1, v_2, \dots, v_k \in V$ form a Jordan chain of T corresponding to the eigenvalue λ . That is,

$$(T - \lambda I)v_1 = 0$$

with $v_1 \neq 0$, and for $i = 2, \dots, k$, the following holds:

$$(T - \lambda I)v_i = v_{i-1}$$

Prove that the set of vectors $\{v_1, v_2, \dots, v_k\}$ is linearly independent.

4) *Inverse system.* Suppose

$$H(s) = C(sI - A)^{-1}B + D,$$

where D is square and invertible. You will find a linear system with transfer function $H(s)^{-1}$.

a) Start with

$$\dot{x} = Ax + Bu, \quad y = Cx + Du,$$

and solve for $\dot{x}$ and u in terms of x and y . Your answer will have the form:

$$\dot{x} = Ex + Fy, \quad u = Gx + Hy.$$

Interpret the result as a linear system with state x , input y , and output u .

b) Verify that

$$(G(sI - E)^{-1}F + H)(C(sI - A)^{-1}B + D) = I.$$

Hint: use the following “resolvent identity”:

$$(sI - X)^{-1} - (sI - Y)^{-1} = (sI - X)^{-1}(X - Y)(sI - Y)^{-1}$$

which can be easily verified by multiplying by $sI - X$ on the left and $sI - Y$ on the right.